

WEEK 2 ASSIGNMENT

BLACK-SCHOLES PRICING MEMO ON A REAL NSE OPTIONS CHAIN

1. UNDERLYING SELECTION AND OPTION CHAIN

For this assignment, Reliance Industries Limited (RIL) was selected as the underlying security. Data was collected from the NSE option chain on 9 June 2026. The selected expiry was the near-month expiry of 30 June 2026.

The spot price of Reliance at the time of observation was ₹1272.30. The strike selected was ₹1270 because it is the closest available strike to the prevailing market price and therefore represents an at-the-money (ATM) option.

Search by Company name, Index or Symbol

23,242.10
119.10 (0.52%)
09-Jun-2026 15:30 IST
Streaming

Futures 30-Jun-2026
23,201.00 -102.00 (-0.44%)
09-Jun-2026 23:42 IST

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RELIANCE1,272.30▲9.00(0.71%)●Close 1,269.20 5.90(0.47%)

▲	660	131	1,307	26.61	62.35	4.00	500	62.25	62.75	500	1,230.00	1,000	11.60	11.70	500	-5.25	11.65	25.23	4,302	262
▲	790	230	1,460	26.43	55.20	4.20	1,000	55.10	55.55	500	1,240.00	500	14.40	14.55	2,000	-5.80	14.45	25.16	5,319	301
▲	2,279	391	3,021	26.49	48.75	3.55	1,000	48.50	48.75	4,500	1,250.00	3,000	17.60	17.75	500	-6.30	17.80	25.17	5,972	-95
▲	1,687	618	7,829	26.27	42.45	3.10	500	42.30	42.45	500	1,260.00	3,000	21.40	21.50	9,500	-7.05	21.45	25.01	7,653	411
▲	2,524	719	10,683	26.20	36.80	2.50	500	36.75	36.80	6,500	1,270.00	500	25.65	25.90	1,000	-7.60	25.70	24.95	6,446	729
▲	4,985	957	11,338	26.10	31.60	2.10	500	31.55	31.60	1,500	1,280.00	500	30.50	30.70	1,000	-7.80	30.50	24.91	4,728	606
▲	2,435	455	4,540	26.05	26.95	1.90	1,000	26.85	26.95	500	1,290.00	500	35.70	36.10	500	-7.90	35.95	24.97	1,801	-11
▲	13,614	936	17,078	26.09	22.90	1.75	3,000	22.80	22.90	1,500	1,300.00	500	41.55	41.85	500	-8.15	41.90	25.04	2,875	-219
▲	3,783	89	5,123	25.94	19.10	1.40	1,500	18.90	19.10	2,000	1,310.00	1,000	47.70	48.25	500	-8.90	47.60	24.46	383	-24
▲	8,952	1,060	9,214	25.78	15.75	1.05	2,500	15.60	15.75	1,500	1,320.00	1,000	54.40	54.80	500	-8.50	54.70	24.70	739	-142
▲	4,999	-90	7,801	25.93	13.15	0.70	4,000	12.95	13.15	3,000	1,330.00	500	61.60	62.10	500	-9.20	61.65	24.37	382	-178

Note:

10% interest rate is applied while computing implied volatility.

Highlighted options are in-the-money.

Volume and Open Interest are displayed in contracts.

Table 1: Option Details

Variable	Value
Underlying	Reliance Industries Ltd
Spot Price (S)	₹1272.30
Strike Price (K)	₹1270
Expiry Date	30-Jun-2026
Days to Expiry	21
Time to Expiry (T)	21/365 = 0.0575 years

Market Call Price ₹36.80

Market Put Price ₹25.70

Call Implied Volatility 26.20%

Put Implied Volatility 24.95%

The strike of ₹1270 was selected because it is closest to the spot price of ₹1272.30 and therefore provides the most appropriate ATM comparison for Black-Scholes valuation.

2. ESTIMATION OF HISTORICAL VOLATILITY (σ)

Historical volatility was estimated using Reliance's recent daily closing prices.

The daily logarithmic return was calculated as:

$$r_t = \ln(P_t/P_{t-1})$$

where:

P_t = Closing price on day t

P_{t-1} = Closing price on the previous trading day

DAT E	SERI ES	OPEN	HIGH	LOW	PREV. CLOS E	LTP	CLOS E	VWAP	52W H	52W L	VOLUME	VALUE	NO. OF TRADES
09-Jun-2026	EQ	1,269.00	1,274.20	1,257.50	1,263.30	1,272.30	1,269.20	1,264.75	1,611.80	1,257.50	2,36,20,214	29,87,37,31,086.30	2,73,651
08-Jun-2026	EQ	1,277.00	1,282.60	1,259.20	1,291.00	1,262.80	1,263.30	1,271.90	1,611.80	1,259.20	1,64,94,759	20,97,96,89,703.10	3,03,591
05-Jun-2026	EQ	1,304.50	1,306.00	1,288.00	1,303.70	1,291.40	1,291.00	1,294.80	1,611.80	1,288.00	1,77,85,223	23,02,83,93,410.00	3,56,802
04-Jun-2026	EQ	1,301.00	1,311.20	1,293.10	1,313.20	1,308.00	1,303.70	1,303.16	1,611.80	1,290.00	2,34,74,327	30,59,07,35,365.60	3,18,716
03-Jun-2026	EQ	1,315.00	1,324.00	1,301.00	1,314.60	1,312.60	1,313.20	1,309.61	1,611.80	1,290.00	2,00,12,293	26,20,82,26,472.80	3,80,780
02-Jun-2026	EQ	1,307.70	1,327.80	1,300.30	1,320.00	1,321.00	1,314.60	1,313.43	1,611.80	1,290.00	2,33,02,785	30,60,65,99,536.00	3,62,323

01-Jun-2026	EQ	1,332.50	1,335.50	1,318.50	1,321.20	1,320.20	1,320.00	1,324.32	1,611.80	1,290.00	1,07,00,603	14,17,09,74,474.50	2,12,031
29-May-2026	EQ	1,364.00	1,369.00	1,317.00	1,350.50	1,321.90	1,321.20	1,330.58	1,611.80	1,290.00	4,26,34,247	56,72,84,19,201.10	3,09,971
27-May-2026	EQ	1,353.00	1,361.00	1,348.60	1,356.30	1,352.00	1,350.50	1,354.62	1,611.80	1,290.00	1,32,15,138	17,90,14,99,784.80	2,30,257
26-May-2026	EQ	1,360.10	1,368.50	1,352.40	1,367.00	1,357.00	1,356.30	1,359.22	1,611.80	1,290.00	1,37,69,747	18,71,61,37,070.80	1,48,444
25-May-2026	EQ	1,366.80	1,371.10	1,357.00	1,354.50	1,366.40	1,367.00	1,363.11	1,611.80	1,290.00	67,50,071	9,20,10,71,110.90	1,39,554
22-May-2026	EQ	1,350.00	1,367.40	1,349.10	1,349.60	1,358.00	1,354.50	1,357.24	1,611.80	1,290.00	71,05,813	9,64,43,25,563.80	1,63,887
21-May-2026	EQ	1,367.20	1,370.90	1,345.20	1,359.70	1,349.00	1,349.60	1,352.83	1,611.80	1,290.00	1,71,50,630	23,20,18,23,031.70	2,75,605
20-May-2026	EQ	1,318.70	1,362.90	1,312.60	1,322.70	1,360.30	1,359.70	1,345.23	1,611.80	1,290.00	1,32,48,515	17,82,23,10,858.20	2,27,919
19-May-2026	EQ	1,338.70	1,344.00	1,318.40	1,335.90	1,322.80	1,322.70	1,332.15	1,611.80	1,290.00	2,16,65,501	28,86,16,86,347.30	3,39,757
18-May-2026	EQ	1,334.00	1,342.00	1,318.70	1,336.40	1,335.00	1,335.90	1,333.44	1,611.80	1,290.00	1,30,22,473	17,36,47,34,585.60	2,24,065
15-May-2026	EQ	1,356.80	1,364.80	1,329.20	1,361.80	1,339.00	1,336.40	1,339.37	1,611.80	1,290.00	1,99,76,192	26,75,55,67,751.40	2,76,450
14-May-2026	EQ	1,365.20	1,378.00	1,358.40	1,358.80	1,368.50	1,361.80	1,365.93	1,611.80	1,290.00	1,73,03,059	23,63,46,93,081.40	3,12,686
13-May-2026	EQ	1,361.40	1,372.40	1,352.40	1,364.00	1,361.20	1,358.80	1,360.83	1,611.80	1,290.00	1,37,97,989	18,77,67,83,797.90	2,53,418
12-May-2026	EQ	1,392.00	1,393.50	1,360.30	1,388.20	1,366.50	1,364.00	1,375.76	1,611.80	1,290.00	2,43,57,500	33,50,99,95,739.50	4,05,245
11-May-2026	EQ	1,420.00	1,428.00	1,382.00	1,435.20	1,385.30	1,388.20	1,402.78	1,611.80	1,290.00	1,52,61,787	21,40,89,32,295.60	2,40,591

The standard deviation of daily log returns was computed and annualized using:

$$\sigma_{\text{annual}} = \sigma_{\text{daily}} \times \sqrt{252}$$

The resulting annualized historical volatility was:

$$\text{Historical Volatility} = 18.27\%$$

The option chain reported an implied volatility of:

$$\text{Implied Volatility} = 26.20\%$$

Table 2: Volatility Comparison

Measure	Value
Historical Volatility	18.27%
Implied Volatility	26.20%
Difference	7.93%

Discussion

The implied volatility exceeds the historical volatility by approximately 7.93 percentage points. This suggests that market participants expect future volatility to be higher than recent realized volatility. Such a difference often reflects uncertainty regarding future events, earnings expectations, macroeconomic developments, or overall market sentiment.

3. RISK-FREE RATE

The RBI repo rate of 6.00% was used as the risk-free rate proxy because it represents the benchmark policy interest rate in India and is widely used in academic and practical valuation exercises.

Quoted Annual Rate:

$$R = 6.00\% = 0.06$$

The continuously compounded equivalent rate is:

$$r = \ln(1 + R)$$

$$r = \ln(1.06)$$

$$r = 0.0583$$

Therefore,

Continuously Compounded Risk-Free Rate = 5.83%

4. BLACK-SCHOLES VALUATION

Inputs

$$S = 1272.30$$

$$K = 1270$$

$$\sigma = 26.20\% = 0.262$$

$$r = 0.0583$$

$$T = 0.0575$$

Step 1: Calculate d_1

$$d_1 = [\ln(S/K) + (r + \sigma^2/2)T] / [\sigma\sqrt{T}]$$

$$d_1 = [\ln(1272.30/1270) + (0.0583 + 0.262^2/2)(0.0575)] / [0.262\sqrt{0.0575}]$$

$$d_1 \approx 0.115$$

Step 2: Calculate d_2

$$d_2 = d_1 - \sigma\sqrt{T}$$

$$d_2 = 0.115 - (0.262 \times \sqrt{0.0575})$$

$$d_2 \approx 0.052$$

Step 3: Calculate $N(d_1)$ and $N(d_2)$

$$N(d_1) \approx 0.546$$

$$N(d_2) \approx 0.521$$

Step 4: Calculate Theoretical Call Price

$$C = SN(d_1) - Ke^{-rT}N(d_2)$$

$$C = (1272.30 \times 0.546) - (1270 \times e^{(-0.0583 \times 0.0575)} \times 0.521)$$

$$C \approx ₹33.15$$

Step 5: Calculate Theoretical Put Price

$$P = Ke^{-rT}N(-d_2) - SN(-d_1)$$

$$P \approx ₹28.05$$

Therefore,

Theoretical Call Price = ₹33.15

Theoretical Put Price = ₹28.05

5. COMPARISON WITH MARKET PRICES

Table 3: Market vs Black-Scholes

Option	Market Price	Black-Scholes Price	Difference
Call	₹36.80	₹33.15	₹3.65
Put	₹25.70	₹28.05	-₹2.35

Discussion

The market call price is slightly higher than the Black-Scholes theoretical value, suggesting that traders are willing to pay a premium for upside exposure. Conversely, the market put price is somewhat below the theoretical estimate.

The differences are relatively small and indicate that the option market is broadly consistent with Black-Scholes pricing. However, deviations may arise due to bid-ask spreads, liquidity considerations, volatility skew, transaction costs, and market expectations not captured by the model.

Black-Scholes assumes constant volatility, while actual option markets exhibit volatility smiles and skews. Therefore, identical volatility assumptions across strikes may not accurately reflect observed market prices.

6. GREEKS ANALYSIS

Delta (Δ)

$$\Delta = N(d_1)$$

$$\Delta = 0.546$$

Interpretation:

A ₹1 increase in Reliance stock price is expected to increase the call option value by approximately ₹0.55.

Gamma (Γ)

$$\Gamma = \phi(d_1)/(S\sigma\sqrt{T})$$

$$\Gamma \approx 0.0050$$

Interpretation:

Gamma measures the rate of change of Delta. A higher Gamma implies greater sensitivity to movements in the underlying stock price.

Vega (v)

$$v = S\phi(d_1)\sqrt{T}$$

$$v \approx 121.6$$

Interpretation:

For a 1.00 increase in volatility (100 percentage points), the option value increases by approximately ₹121.6. Therefore, a 1% increase in volatility increases option value by roughly ₹1.22.

Theta (Θ)

$$\Theta \approx -0.70 \text{ per day}$$

Interpretation:

The option loses approximately ₹0.70 in value per day due to time decay, assuming all other variables remain unchanged.

Table 4: Greeks Summary

Greek	Value	Interpretation
Delta	0.546	Price sensitivity
Gamma	0.0050	Change in Delta
a		
Vega	121.6	Volatility sensitivity
Theta	-0.70/day	Time decay

Scenario Analysis

Scenario 1: Underlying rises by 1%

$$\Delta S = 1\% \times 1272.30$$

$$\Delta S = 12.72$$

Predicted option price change:

$$\Delta C \approx \text{Delta} \times \Delta S$$

$$\Delta C \approx 0.546 \times 12.72$$

$$\Delta C \approx ₹6.95$$

Predicted Call Price:

$$₹36.80 + ₹6.95$$

$$\approx ₹43.75$$

Scenario 2: Volatility rises by 1%

$$\Delta \sigma = 0.01$$

$$\Delta C \approx \text{Vega} \times \Delta \sigma$$

$$\Delta C \approx 121.6 \times 0.01$$

$$\Delta C \approx ₹1.22$$

Predicted Call Price:

$$₹36.80 + ₹1.22$$

$$\approx ₹38.02$$

7. AMERICAN-STYLE ADJUSTMENT

Black-Scholes assumes European exercise, meaning options can only be exercised at expiry. Many NSE stock options are American-style and permit early exercise.

For the selected Reliance option, the early exercise premium is likely to be small. The option is near at-the-money and has a relatively short time remaining until expiry. Exercising early would sacrifice the remaining time value embedded in the option.

Early exercise becomes more attractive primarily for deep in-the-money options or when significant dividend payments are expected before expiry. Since neither condition is especially relevant for the selected option, Black-Scholes remains a reasonable approximation.

Therefore, the early-exercise premium is expected to be small and does not materially affect valuation.

8. REFLECTION

Applying Black-Scholes to an actual NSE option chain highlighted the difference between theoretical models and real market behaviour. The most surprising observation was the gap between historical volatility and implied volatility. While recent price movements suggested annualized volatility of only 18.27%, the market was pricing volatility at over 26%. This demonstrated that option prices reflect expectations about the future rather than merely past price behaviour. Another important insight was that even when Black-Scholes assumptions

are not perfectly satisfied, the model still provides reasonably close estimates of market prices. Working through the calculations also showed how sensitive option prices are to volatility assumptions. Small changes in volatility produced noticeable changes in theoretical values and Greeks. Finally, observing actual market prices reinforced the importance of volatility smiles, liquidity effects, and trader expectations, factors that are often discussed theoretically but become much clearer when analysing a live option chain.

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